5.Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

Aim

To write a Python function to analyze and display the histogram of an input image based on its color levels using OpenCV.

Procedure

1. Import OpenCV and Matplotlib libraries.
2. Read the input image.
3. Check whether the image is loaded successfully.
4. Calculate the histograms for Blue, Green, and Red color channels.
5. Display the histograms using Matplotlib.
6. Observe the distribution of color levels.

Code

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("img1.jpg")

if img is None:
    print("Image not found")
else:
    colors = ('b', 'g', 'r')

    for i, color in enumerate(colors):
        hist = cv2.calcHist([img], [i], None, [256], [0, 256])
        plt.plot(hist, color=color)

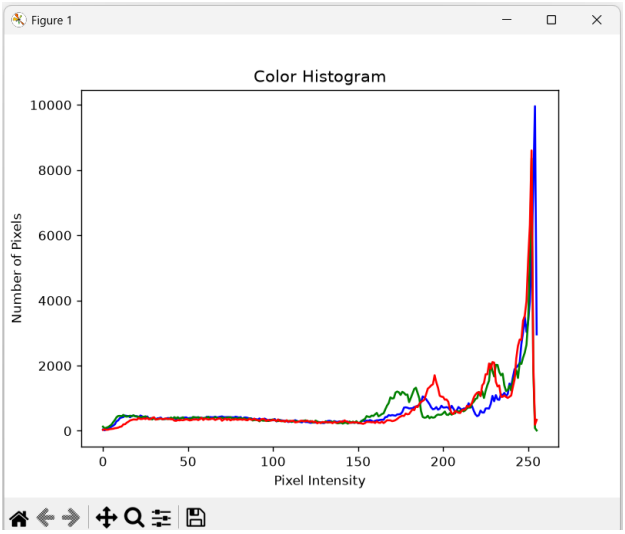
    plt.title("Color Histogram")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Number of Pixels")
```

```
plt.show()
```

input:



output:



Result

Thus, the histogram of the input image was successfully analyzed and displayed for the Blue, Green, and Red color channels using OpenCV.